

Chapter 16 worked solutions – Continuous probability distributions

Solutions to Exercise 16G

1a $\mu = 7.5$ and $\sigma = 2.5$

1b

Class centre	0.5	1.5	2.5	3.5	4.5	5.5	6.5	7.5
frequency	2	5	24	46	88	116	138	158
Relative frequency	2	7	31	77	165	281	419	577

Class centre	8.5	9.5	10.5	11.5	12.5	13.5	14.5
frequency	144	113	78	47	27	10	4
Relative frequency	721	834	912	959	986	996	1000

$$E(X) = \sum xP(x) = \frac{0.5 \times 2 + 1.5 \times 5 + 2.5 \times 24 + \dots + 14.5 \times 4}{1000} = \frac{7533}{1000} = 7.533 \div 7.5$$

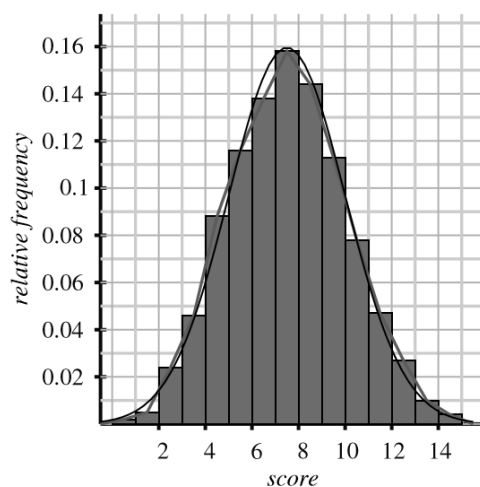
$$\begin{aligned} \text{Var}(X) &= E(X^2) - E(X)^2 \\ &= \sum x^2 P(x) - E(X)^2 \\ &= \frac{63\,034}{1000} - 7.533^2 \\ &= 6.287\,911 \end{aligned}$$

$$\text{Therefore, } \sigma = \sqrt{\text{Var}(X)} = \sqrt{6.287\,911} = 2.50757 \dots \div 2.5$$

Yes, the answers agree with the previous results.

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1c



1d Either perform the experiment more than 1000 times, or average more than three random numbers at each stage.

4c The mean should be about 5 and the standard deviation about 1.6.

8d The maximum point on the curve has coordinates $\left(0, \frac{1}{\sqrt{2\pi}}\right)$.